Yufeng (Alex) Xia

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Research Interests

- **Machine learning systems** and **AI-centric networking** — how an AI workload shares accelerators, and the network between them, with a latency-critical tenant; **high-performance data planes** (event-driven transport, zero-copy, pinned memory, serialisation cost); and **large-scale emulation** of both.
- **Availability:** MSc by Research thesis submits September 2026. Available remotely from **September 2026**, full-time (remote or in Hong Kong) from **October 2026**. Applying for **PhD entry in Fall 2027** and to the **Hong Kong PhD Fellowship Scheme**.

Education

- **University of Edinburgh — MSc by Research, Computer Science** Edinburgh, UK
  Supervised by Prof. Mahesh K. Marina. Thesis: Flux-DT: An Elastic, Preemption-Tolerant Digital Twin Runtime for AI-RAN *Sep 2025 – Sep 2026 (expected)*
  - Research-only degree (thesis, no taught coursework); sole author of the thesis and its implementation.
- **University of Edinburgh — BSc (Hons) Computer Science** Edinburgh, UK
  *Grade: **First Class Honours** (weighted average 74/100; UK First Class ≥ 70/100)* *Sep 2021 – May 2025*
  - **70+** in *Machine Learning Systems*, *Machine Learning Practical*, and *Computer Communications and Networks.*

Selected Publications

EmuRAN: Scalable and Cost-Effective Cloud-based RAN Emulation System *MobiCom'26 (cond. accepted)*
*Ujjwal Pawar, Andrew E. Ferguson, **Yufeng Xia**, Xenofon Foukas, Mahesh Marina, Bozidar Radunovic*

Morphling: Emulator for Distributed Machine Learning at the Edge *EdgeSys'26 (Best Paper)*
*Leyang Xue, **Yufeng Xia**, Eren Mendi, Ismaeel Bashir, Jiaxun Yang, Myungjin Lee, Mahesh Marina*

Weaver: Foundation Model Training over AI-RAN Compute Infrastructure *under submission*
*Leyang Xue, Tianxin Wang, Xin Zhe Khooi, Jiaxun Yang, Dheeraj Mahendiran, **Yufeng Xia**, Mun Choon Chan, Myungjin Lee, Mahesh Marina*

Wasp: Foundation Model Training System for the Edge *SEC'26*
*Leyang Xue, **Yufeng Xia**, Myungjin Lee, Mahesh Marina*

CoSenseAQ: Compiler Optimizations using Sensor Technical Specifications with Automatic Quantization *under submission*
***Yufeng Xia**, Pei Mu, Antonio Barbalace*

Research and Project Experience

- **Weaver — Foundation Model Training over Shared AI Compute Infrastructure** Under Submission
  Research Assistant; GPU profiling, interference characterisation, controller evaluation *2024 – 2026*
  - **Harvesting accelerators from a latency-critical tenant**: Evaluated the controller that decides how much accelerator capacity an AI workload may harvest from infrastructure shared with a latency-critical tenant, a near-real-time resource-allocation problem on GPU-accelerated hardware. On a multi-site testbed it harvests up to **83%** of the available spare compute without degrading the co-tenant's performance.
  - **Roofline analysis and deadline measurement**: Ran the GPU profiling and roofline analysis showing the priority workload's kernels are memory-bandwidth-bound while training is compute-bound — the asymmetry that makes SM-level co-location possible; instrumented its scheduler to measure per-deadline completion, and carried out the large-scale distributed training runs used in the evaluation.
- **Morphling — Emulator for Distributed Machine Learning at the Edge** EdgeSys 2026
  *Research Assistant; **Best Paper Award**; owned the C++ transport data plane* *Oct 2025 – Mar 2026*

- **Inter-device transport**: Built the event-driven (libevent) proxy client/server carrying tensor traffic between emulated devices — **2,400 lines of C++** in an 18k-line C++/CUDA codebase. Emulation fidelity is bounded by this path, so transfer cost had to be driven to the microsecond level.
- **Zero-copy and memory-path optimisation**: Designed a bucketed `cudaHostAlloc` pinned-memory pool and zero-copy scatter-gather buffers to remove per-transfer allocation and staging copies; added CUDA host registration and thread pinning, each validated with dedicated microbenchmarks.
- **Serialisation cost and distributed correctness**: Measured serialisation overhead across Protobuf and FlatBuffers along the full send path and chose the batching strategy from the measurements; implemented virtual-clock synchronisation across emulated devices and fixed the connection-cleanup and device-failure deadlocks that appeared only at scale.

- **Flux-DT — Elastic, Preemption-Tolerant Runtime for Stateful Workloads on Reclaimable Compute** **MSc by Research Thesis**
  University of Edinburgh — sole author *Sep 2025 – Present*
  - **State that must survive preemption**: Existing compute-harvesting designs assume the harvesting workload is stateless and disposable. Built a runtime for a workload that is neither: it holds state that must survive interruption, so it can run on bursty, reclaimable capacity without violating the priority tenant's guarantees or corrupting its own state.
  - **Near-real-time control policy**: Designed and implemented a control loop that answers resource pressure with a graded set of actions — relocation, fidelity reduction, suspend and resume — rather than a single eviction decision, on the timescale the priority workload imposes.
  - **Scheduling heuristic and shedding order**: Formulated work placement as a joint objective across interfering units and **designed a greedy heuristic** to decide which work is worth doing; the same ranking determines the shedding order when compute is reclaimed mid-solve.
- **EmuRAN — Scalable, Cost-Effective Cloud-based RAN Emulation System** MobiCom 2026
  Research Assistant; ***owned the end-to-end evaluation*** *Conditionally Accepted*
  - **Cloud deployment at scale**: Deployed and operated the emulator on public cloud across **130 instances and 6,400 CPU cores**, emulating 10,000 mobile devices and 10,000 base stations, two orders of magnitude larger than existing software solutions.
  - **Fidelity and scalability evaluation**: Designed and ran the evaluation campaign: fidelity against real hardware (random-access timelines, iperf3 throughput, per-slot completion-time distributions) and scalability under node count and constrained compute, establishing the system behaves correctly rather than merely running at scale.
- **CoSenseAQ — LLVM Auto-Quantization from Sensor Specifications** Under Submission
  Research Assistant, ICSA; ***first author*** *Apr 2024 – May 2025*
  - **Compiler-level optimisation**: Built an LLVM auto-quantization framework driven by information a compiler cannot recover from source alone, using variable-range analysis and type shrinkage to reduce code size and execution time on embedded targets; low-level C/C++ and cross-compilation toolchain work throughout.
  - **Measurement on real hardware**: Evaluated on ARM Cortex-M0 and Cortex-A53, measuring power, performance and area. Drove the work from problem formulation through implementation and evaluation to manuscript.

Technical Skills

- **Languages** C++ (primary; C++17, the language of all my systems work), C, Python, Java, Bash.
- **High-Performance Data Plane** Event-driven asynchronous I/O (libevent), zero-copy scatter-gather buffers, pinned-memory pooling and CUDA host registration, serialisation cost analysis (Protobuf, FlatBuffers), batching and framing, connection lifecycle management, thread pinning, microsecond-level latency instrumentation.
- **GPU and ML Systems** CUDA programming and profiling (Nsight), SM-level partitioning via green contexts and MPS, cuBLAS/cuBLASXt, roofline and arithmetic-intensity analysis, PyTorch, distributed foundation-model training, scheduling under contention with a latency-critical co-tenant.
- **Distributed and Cloud** Multi-site deployment on Microsoft Azure and CloudLab (130+ instances, 6,400 cores), Kubernetes, Docker, distributed virtual-time synchronisation, failure and deadlock diagnosis at scale.
- **Compilers** LLVM (IR-level passes, variable-range analysis, automatic quantization), MLIR (own multi-target tensor compiler with operator fusion, NVVM/ROCDL lowering to PTX/cubin), cross-compilation for embedded ARM.

- **Networking and Wireless** 5G RAN experimentation with OpenAirInterface and Open5GS, RAN emulation and network digital twins, throughput/latency measurement (iperf3, packet-level tracing).
- **Engineering** Git and code review, CMake, GitHub Actions CI, pre-commit, CodeQL, containerised reproducible builds, technical documentation (see attached toolchain and repository summaries).

Industry Experience

- **Senior Tech — Back-end Development Intern** Suzhou, China
  Production backend APIs in a commercial engineering team; live platform migration from JFinal to Spring Boot, code review and release testing. Java, MySQL, Redis, Kafka. *Jun 2023 – Sep 2023*

Awards, Test Scores and Service

- **Award Best Paper**, EdgeSys 2026 (9th Workshop on Edge Systems, Analytics and Networking, co-located with ACM MobiSys 2026), for *Morphling*.
- **Test Scores GRE 336/340** (Verbal 166, Quantitative 170/170, Jun 2024); IELTS 7.0.
- **Service** Student volunteer, EuroSys 2026 (on-site conference operations).
- **Languages** Mandarin (native), English (fluent).

References

- **Prof. Mahesh K. Marina** — MSc by Research supervisor, School of Informatics, University of Edinburgh · mahesh@ed.ac.uk
- **Dr Antonio Barbalace** — BSc dissertation supervisor and Junior Research Assistant supervisor, School of Informatics, University of Edinburgh · antonio.barbalace@ed.ac.uk